

The Three-Dimensional Spherical Quantum Harmonic Oscillator: A Theoretical Exploration

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ABSTRACT

- The three-dimensional quantum harmonic oscillator is analyzed in spherical coordinates.
- The Schrödinger equation is solved using separation of variables.
- The system exhibits a quantized energy spectrum determined by quantum numbers.
- Results describe the structure of radial and angular wavefunctions.

INTRODUCTION & BACKGROUND

- The quantum harmonic oscillator is a fundamental model in quantum mechanics.
- The time-independent 3-D Schrödinger equation in Cartesian coordinates is given by

$$\frac{-\hbar^2}{2m} \nabla^2 \Psi + V(x, y, z) \Psi = E \Psi.$$

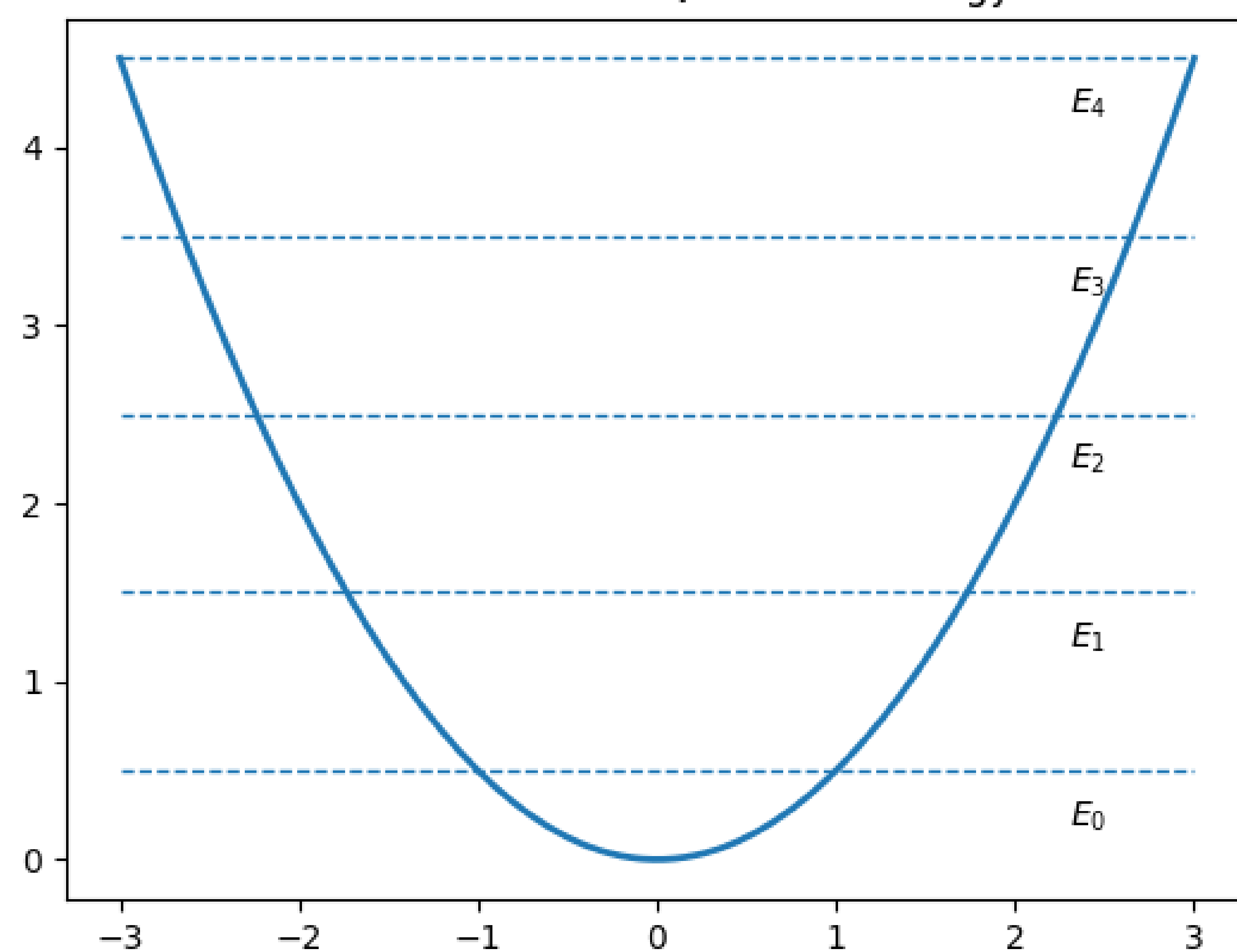
- The potential is given by

$$V(x, y, z) = \frac{1}{2} m \omega^2 (x^2 + y^2 + z^2)$$

where ω is angular frequency and m is the particle's mass.

- In 3-D spherical symmetry, spherical coordinates are most natural.
- Solving the Schrödinger equation leads to discrete energy levels.

Harmonic Oscillator with Quantized Energy Levels



REFERENCES

Brandt et al., *Special Functions of Mathematical Physics*, Springer (2003).
Yan & Huang, *Visualization and Analysis of 3D Linear Harmonic Oscillator*, AJMS (2023).

METHODOLOGY

- Begin with the time-independent Schrödinger equation.
- Expressing the Laplacian $\nabla^2\Psi$ in spherical coordinates yields:

$$\frac{-\hbar^2}{2m} \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Psi}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \Psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 \Psi}{\partial \phi^2} \right) + V(r)\Psi = E\Psi$$
 with potential $V(r) = \frac{1}{2} m\omega^2 r^2$.
- Apply separation of variables to decouple radial and angular differential equations and solve each part individually.

RESULTS

- Separation of variables yields a solution of the form

$$\Psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$$

with angular and radial solutions

$$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}$$

$$\theta(\theta) \propto P_{\ell}^{|m|}(\cos \theta)$$

$$R(r) = N_k r^\ell e^{\frac{-r^2}{2a^2}} L_k^{\ell+\frac{1}{2}}\left(\frac{r^2}{a^2}\right)$$

where the normalization factor $N_k = \sqrt{\frac{2k!}{a^3 \Gamma(k+\ell+\frac{3}{2})}}$

and associated Laguerre polynomials are defined as

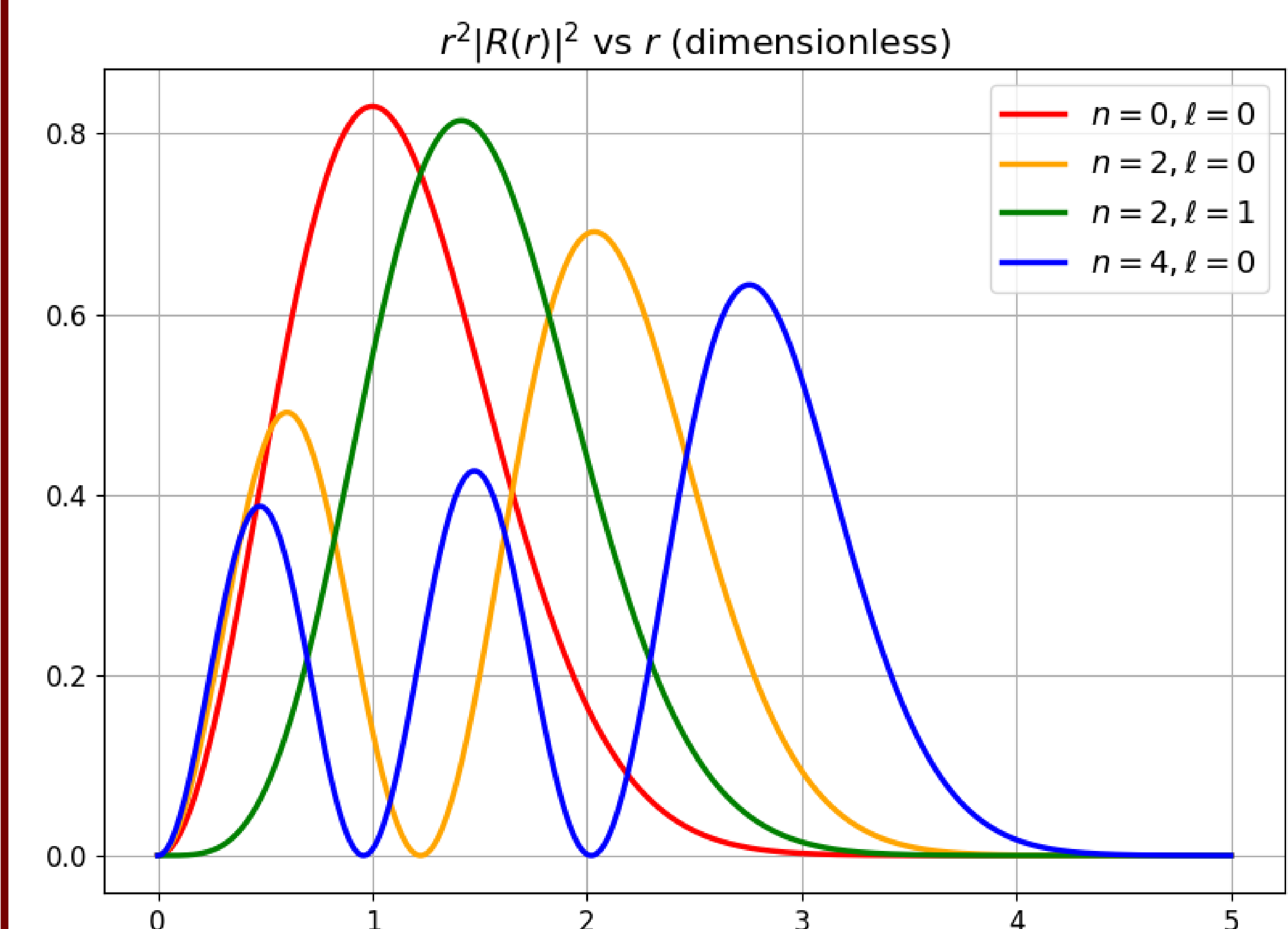
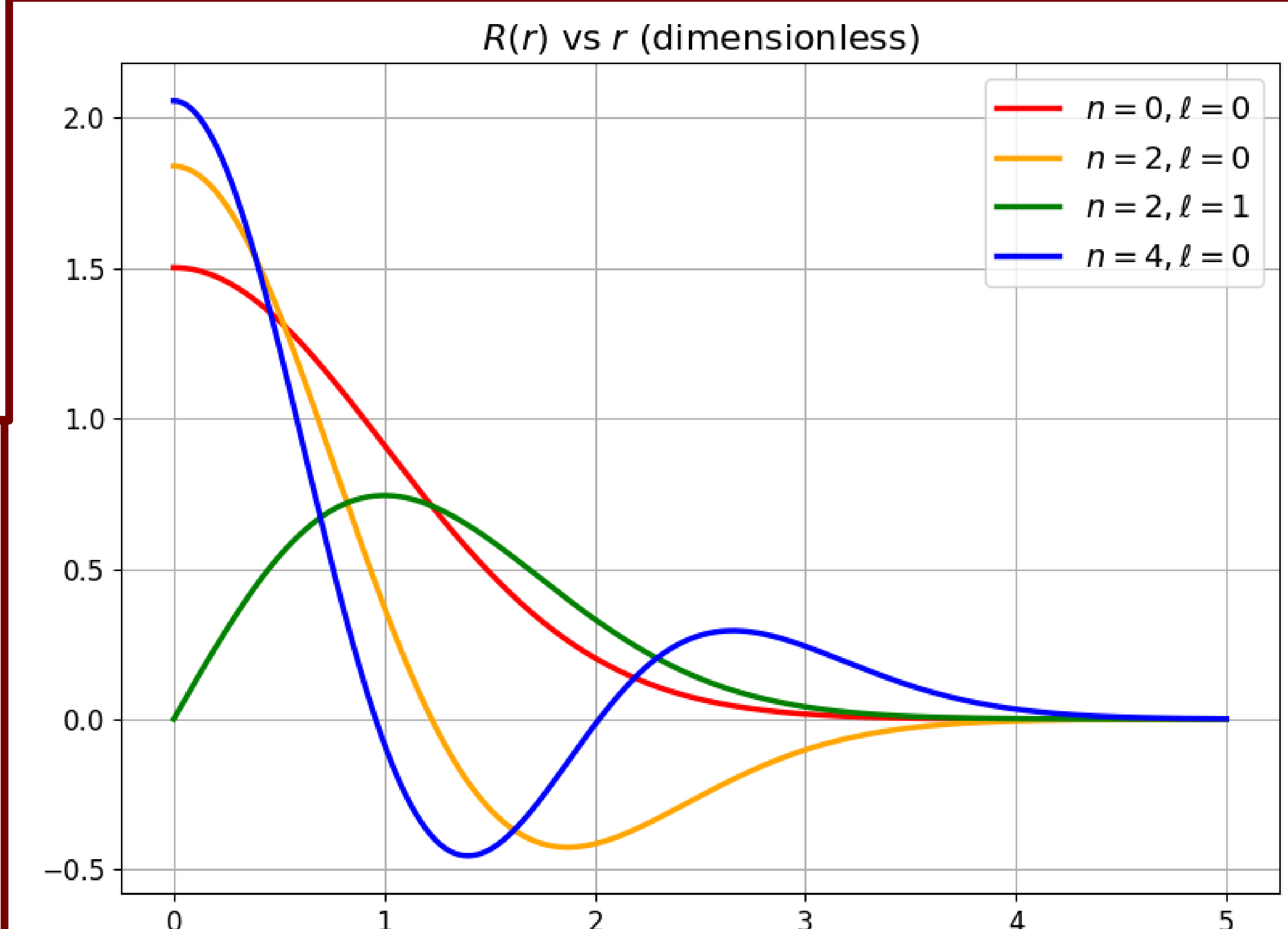
k	$L_k(x)$ with $x = \frac{r^2}{a^2}$
0	-1
1	$-x + 1$
2	$\frac{1}{2}(x^2 - 4x + 2)$

- Quantum numbers arise from boundary conditions and define the energy, angular momentum, and orientation of each state:

Magnetic (m)	$m \in \mathbb{Z}, \quad m \leq \ell$
Angular momentum (ℓ)	$\ell = n, n-2, n-4, \dots \geq 0$
Radial (k)	$k = \frac{n-\ell}{2}$
Total (n)	$n = 2k + \ell, \quad n = 0, 1, 2, \dots$

- Energy quantization arises from requiring normalizable solutions to the radial equation:

$$E_n = \hbar\omega(n + \frac{3}{2}).$$



Plots of radial wavefunctions (top) and their probability densities (bottom) created via Python.

CONCLUSIONS

- $\Theta(\theta)\Phi(\phi)$ determines the wavefunction's shape & orientation in space.
- $R(r)$ determines how the wavefunction varies with distance from the origin.
- Quantum numbers define energy, angular momentum, & structure.
- Higher energy states have more nodes and larger spatial extent.
- Radial probability shifts outward with increasing energy.